



#### GLASS JELLIES

Glass jellyfish live in all the world's oceans. They are found in the sunlit upper zone as well as at depths of 2,300 ft (700 m). They have a deep bell with a long mouth hanging down, which twists round to catch tiny prey. Glass jellyfish can put on displays of beautiful rainbow colors.

Sharply angled dorsal fin

Silvery body helps camouflage fish in right light

Large eye helps spot prey in the dark

Luminous patches in the mouth can attract prey right inside the jaw

Upward-pointing eye helps locate prey

Model of a hatchet fish, *Opisthoproctus*

#### LOOKING UP

There is more than one type of hatchet fish. This bizarre-looking hatchet fish has large, tubular eyes pointing upward so it can see its prey, such as deep-water siphonophores (jellyfishlike relatives of sea firs). The welldeveloped eyes can detect the faintest glimmer of their prey's light. This fish is found in oceans around the world below regions with warm surface waters.

Extremely tiny second dorsal fin is fleshy

Lancet fish can grow to 6 ft 6 in (2 m) long

Colors (instead of light organs) help fish blend into light ocean above and dark ocean below

Model of a hatchet fish, *Sternoptyx*

Light organs also found under tail

#### DEEP-SEA HATCHET

Hatchet fish have bladelike, silvery bodies. The light organs along their belly and tail, when viewed from below, help them blend in with the brighter surface water above. These must be the right degree of brightness so as not to outshine the light from above, which would make them easier to see. Hatchet fish live in the Atlantic, Pacific, and Indian Oceans.

Large, first dorsal fin

Symmetrical tail fin

Tube along belly produces light to avoid being spotted from below by predators

Anus contains a colony of luminescent bacteria that generate light

Asymmetrical caudal (tail) fin

Lancet fish swims below the warm surface waters of the Atlantic, Pacific, and Caribbean